23-24/03/19
UNIVERSITY OF CAPE COAST
COLLEGE OF DISTANCE EDUCATION
EXAMINATIONS UNIT

UNIVERSITY OF CAPE COAST
COLLEGE OF DISTANCE EDUCATION
END OF FIRST SEMESTER EXAMINATION
DIPLOMA IN MATHEMATICS/SCIENCE EDUCATION
DSE/DME YR 1, 2018/2019

MAT101: ALGEBRA AND TRIGONOMETRY

EXAM CODE: 01

CANDIDATE'S SIGNATURE .....

REGISTRATION NO: ..... TIME ALLOWED: 2 HOURS

Instructions:

1. **Provide your EXAM CODE at the front page of the answer booklet. Failure to write your EXAM CODE will result in a score of zero (0) for section A.**
2. Answer ALL Questions in Section "A" in the answer booklet.
3. Answer *any two (2)* Questions in Section "B" in the answer booklet provided

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SECTION A *[Answer ALL Questions in this Section]* *(20 MARKS)*

1. How many subsets have the set of integers $A = \{x : -1 < x \leq 2\}$?
   A. 3
   B. 4
   C. 8
   D. 12

2. Given that $A = \{x : -10 < x \leq -1)\}; B = \{x : -10 < x \leq 2\}$ and $C = \{x : -10 < x \leq 1)\}$ are all subsets of the universal set of integers $U = \{x : -10 < x \leq 20\}$ how many members are in $(A \cap B \cap C)$?
   A. 0
   B. 1
   C. 12
   D. 18

3. A = {numbers divisible by 5} B = {odd numbers from 3 to 17 inclusive} and C = {prime numbers less 23}, where A, B and C belong to the set of natural numbers. Find $A \cap B \cap C$.
   A. { }
   B. {5}
   C. {5, 15}
   D. {5, 10, 15}

4. Which of the following is true?

A. $\frac{-7}{8} > \frac{8}{9}$

B. $\frac{-11}{-12} > \frac{-12}{-5}$

C. $\frac{4}{5} > \frac{3}{4}$

D. $\frac{9}{10} > \frac{10}{11}$

5. Which of the following equations has **no** solution in the set of natural numbers?

A. $8x-7=9(x-1)$

B. $5x-1=26$

C. $2(x+3)=3(x-4)$

D. $2(x+1)=5(x-2)$

6. Rationalize: $\frac{2+\sqrt{3}}{2-\sqrt{3}}$

A. $\frac{2-\sqrt{3}}{3-\sqrt{2}}$

B. $\frac{2+\sqrt{3}}{2-\sqrt{3}}$

C. $4+\sqrt{3}$

D. $7+4\sqrt{3}$

7. Which of the following when rationalized leaves an even denominator?

A. $\frac{13}{3-\sqrt{7}}$

B. $\frac{15}{3-\sqrt{2}}$

C. $\frac{17}{13+\sqrt{12}}$

D. $\frac{19}{11-\sqrt{8}}$

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d) Divide x^3-5x+7 7 by $x-1$ **(5 marks)**

3. a) i) Find the inverse of the matrix

$$B=\begin{pmatrix}4 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 4\end{pmatrix}$$

(3 marks)

ii) Write the matrix equation for the system of equations: **(1 mark)**

$$\begin{cases} x+y+z=3 \\ 5y+z=6 \\ 2x+y=3 \end{cases}$$

iii) Hence or otherwise, solve the system. **(6 marks)**

b) A property developer can build at most 10 three-bed flats and at most 15 two-bed flats on a piece of land. Each three-bed flat requires 6 months of labour and each two-bed flat 4 months of labour. The maximum total number of months of labour that the builder has at his disposal is 96.

i) Give **three (3)** inequalities that express the conditions above. **(3 marks)**
ii) Graph and shade the region that satisfies these inequalities. **(4 marks)**
iii) Find the maximum number of three-bed and two-bed flats that the developer can build within the time at his disposal. **(3 marks)**

4. a) Find the **six (6)** trigonometric functions of the angle ϕ formed in the third quadrant and measured anticlockwise from the horizontal axis and the line from the origin to the point P(-8, -12) **(10 marks)**

b) Show that $\frac{1-\cos t}{\sin t}=\frac{\sin t}{1+\cos t}$ **(4 marks)**

c) Verify the identity $\cos^4\theta-\sin^4\theta=1-2\sin^2\theta$ **(2 marks)**

d) Given that $\sin\beta=\frac{12}{13}$, $0\le\beta\le\frac{\pi}{2}$, evaluate $\cos^2(\beta)-\sin^2(\beta)$. **(4 marks)**

SECTION B

Answer ANY TWO (2) Questions from this section.

1. a) Suppose that the set $Q = \{1, 2\}$ $R = \{m\}$, and find

   i. $|Q \times R|$ **(1 mark)**

   ii. the number of relations from R to Q **(1 mark)**

   iii. the number of relations on Q **(1 mark)**

   b) Thirty-six students indicated which of the following languages they studied in the Senior High School: Twi (T), Ga (G) and Dagbani (D). Twenty studied Akan, 17 studied Ga and 14 studied Dagbani. Five students studied Akan and Ga only; 1 studied Akan and Dagbani only while 3 studied Akan and Dagbani only. Eight studied Akan only while 6 studied Dagbani only. Two students did not study any of the three languages.

   i. Illustrate the above information on a Venn diagram **(3 marks)**

   ii. Find the number of students who studied

   (a) two languages **(3 marks)**

   (b) all the three languages **(3 marks)**

   (c) Ga only **(2 marks)**

   c) For any two sets E and F, prove that $(E \cup F)' = E' \cap F'$. **(3 marks)**

   d) Given that Set $M = \{x : -5 < x \leq 5\}$, $N = \{x : -7 \leq x \leq 7\}$ and $P = \{x : -2 < x < 2\}$ are subsets of the universal set of integers, list members of the $M \cap N \cap P$. **(3 marks)**

2. a) If the roots of the equation $3x^2 - 17x + 10 = 0$ are α and β, find the equation whose roots are $\frac{\beta}{\alpha}$ and $\frac{\alpha}{\beta}$. **(5 marks)**

   b) If $f(x) = \frac{x}{10+x}$ and $g(x) = \frac{x}{100-x^2}$, find $f(x) - g(x)$, and state its domain. **(5 marks)**

   c) Simplify $f(x) = \left(\frac{x}{x+8}\right) \div \left(\frac{10}{x^2-64} - \frac{1}{x-8}\right)$ **(5 marks)**

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8. Find the determinant of $B = \begin{pmatrix} 3 & 1 \\ 3 & 1 \end{pmatrix}$.

   A. -8
   B. 0
   C. 2
   D. 4

9. Find the determinant of $D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

   A. -2
   B. 2
   C. 4
   D. 8

10. Find a matrix $C = A \times B$ given: $A = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$

    A. $C = \begin{pmatrix} 4 & 8 \\ 8 & 4 \end{pmatrix}$

    B. $C = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$

    C. $C = \begin{pmatrix} 6 & 1 \\ 1 & 6 \end{pmatrix}$

    D. $C = \begin{pmatrix} 16 & 4 \\ 4 & 16 \end{pmatrix}$

11. Find the inverse of the matrix $M = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$.

    A. $\frac{1}{25}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

    B. $\frac{1}{5}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

    C. $\frac{1}{5}\begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$

    D. $\begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$

12. Which of the following is the solution set of the system of matrix equation $\begin{pmatrix} 2 & 2 & 3 \\ 1 & 2 & 2 \\ 1 & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 14 \\ 10 \\ 2 \end{pmatrix}$.

A. {(-2,-1, 0)}

B. {(1,1,2)}

C. {(2,2,0)}

D. {(2,2,2)}

13. Given that A = {(2, -1), (2, -1), (4, -1), (4, -1), (8, -1)}, how many members have the range of the relation A?

A. 1
B. 3
C. 4
D. 5

14. If the roots of the equation $x^2 - 4x + 1 = 0$ are α and β find the equation whose roots are (i) $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

A. $x^2 - 4x - 1 = 0$
B. $2x^2 - 4x + 1 = 0$
C. $4x^2 - 4x - 1 = 0$
D. $4x^2 - 4x + 1 = 0$

15. If $3^{3x+3} = 27^3$, find x.

A. -1
B. 2
C. 3
D. 9

16. Find the coefficient of x^4 in the binomial expansion of $(3x-5)^4$.

A. 15
B. 60
C. 81
D. 625

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17. A sequence is given by the formula $u_n = 2n - 1$ *for* $n = 1,2,3,\ldots L$. Find the sum of the first 10 terms.

A. 10
B. 11
C. 50
D. 100

18. The sum to infinity of a GP is 2. If the first term is 1, what is the common ratio?

A. $\frac{1}{3}$
B. $\frac{1}{2}$
C. 0.6
D. 1.2

19. Convert 1.5π radians to degrees.

A. 45°
B. 90°
C. 180°
D. 270°

20. Find the size of the angle formed in the first quadrant and measured anticlockwise from the horizontal axis and the line from the origin to the point P(-60, -60).

A. 45°
B. 135°
C. 225°
D. 315°